

## 1. Water Jug Problem(4-Gallon and 3-Gallon Jugs):

### Solution

#### Step-by-Step Solution

##### Step 4-Gallon Jug 3-Gallon Jug Action

|         |   |   |                                                    |
|---------|---|---|----------------------------------------------------|
| Initial | 0 | 0 | Both jugs are empty                                |
| 1       | 0 | 3 | Fill the 3-gallon jug                              |
| 2       | 3 | 0 | Pour 3 gallons into the 4-gallon jug               |
| 3       | 3 | 3 | Fill the 3-gallon jug again                        |
| 4       | 4 | 2 | Pour water into the 4-gallon jug until it is full  |
| 5       | 0 | 2 | Empty the 4-gallon jug                             |
| 6       | 2 | 0 | Pour the remaining 2 gallons into the 4-gallon jug |

### Final Answer

The 4-gallon jug contains exactly 2 gallons, which satisfies the goal.

## 2. Mars Rover – Analyze and Explore the Samples :

### Solution

#### 1. Percepts of the Agent

Percepts are the inputs received by the Mars Rover through its sensors.

- Images from navigation and scientific cameras.
- Soil and rock composition data.
- Temperature and atmospheric pressure.
- Terrain information (rocks, slopes, sand).
- Distance from obstacles.
- Position and direction using navigation sensors.
- Battery and power status.
- Weather conditions such as dust storms and wind.

## 3.8-Queens Problem using Search Space

### Solution

The Backtracking Search Algorithm is commonly used to solve the 8-Queens Problem.

## **Steps**

1. Start with an empty  $8 \times 8$  chessboard.
2. Place the first queen in the first column.
3. Move to the next column and place another queen in a safe position.
4. Before placing a queen, check that no queen attacks it.
5. If no safe position exists, backtrack to the previous column and move the queen to the next possible row.
6. Repeat the process until all eight queens are placed safely.
7. The final board configuration represents the solution.

## **Search Strategy**

- State Space Search
- Backtracking (Depth-First Search)

## **4. OLA Cab Booking – Intelligent Agent**

### **Solution**

The Goal-Based Agent is the most suitable intelligent agent for the OLA Cab Booking system because it makes decisions based on achieving the user's goal of reaching the destination efficiently.

### **Working Steps**

1. The customer opens the OLA application.
2. The customer enters the pickup and destination locations.
3. The customer selects a preferred cab type.
4. The system checks nearby available drivers.
5. The agent calculates the estimated fare and arrival time.
6. The nearest suitable driver is assigned.
7. The booking is confirmed and ride details are displayed.
8. The customer reaches the destination successfully.

## **5. Uniform Cost Search (UCS):**

### **Solution**

### **Uniform Cost Search (UCS)**

Uniform Cost Search (UCS) is a graph search algorithm that expands the node with the smallest accumulated path cost from the start node. It uses a priority queue to store nodes based on their cumulative cost.

### **Steps**

1. Start from the source node **S** with cost **0**.
2. Insert the source node into the priority queue.
3. Remove the node with the lowest cumulative cost.
4. If it is the goal node **G**, stop the search.
5. Otherwise, expand its neighboring nodes.
6. Update the cumulative cost for each neighbor.
7. Add the neighbors to the priority queue.
8. Repeat the process until the goal node is reached.

### **Search Strategy**

- State Space Search
- Uniform Cost Search (Best-First Search based on path cost)